

What rule of inference is used in each of these arguments?

1. Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials.
2. It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous.
3. Linda is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard.
4. Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum.
5. If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material.
6. Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
7. Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
8. If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
9. If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
10. If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.

Use rules of inference to show that the hypotheses “If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on,” “If the sailing race is held, then the trophy will be awarded,” and “The trophy was not awarded” imply the conclusion “It rained.”

Use the following propositional variables in your *argument*:

r : = It Rains, f : = It is Foggy, s : = Sailing race will be held,
 l : = Life saving demonstration will go on, t : = Trophy will be awarded

You are about to leave for school in the morning and discover that you don't have your glasses. You know the following statements are true:

- a. If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.
- b. If my glasses are on the kitchen table, then I saw them at breakfast.
- c. I did not see my glasses at breakfast.
- d. I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.
- e. If I was reading the newspaper in the living room then my glasses are on the coffee table.

Where are the glasses? (*Hint: Use rules of inference.*)

Use the following propositional variables in your *answer*:

RK := I was reading the newspaper in the kitchen.

GK := My glasses are on the kitchen table.

SB := I saw my glasses at breakfast.

RL := I was reading the newspaper in the living room.

GC := My glasses are on the coffee table.

Give 5 examples each of

- *Fallacy of Affirming the Conclusion*
- *Fallacy of Denying the Hypothesis*

For each of these arguments, explain which rules of inference are used for each step.

1. “Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job.”
2. “Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution.”
3. “Each of the 93 students in this class owns a personal computer. Everyone who owns a personal computer can use a word processing program. Therefore, Zeke, a student in this class, can use a word processing program.”
4. “Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean.”

For each of these arguments determine whether the argument is correct or incorrect. If the argument is correct, mention the rule of inference otherwise mention the fallacy. Justify your answer with proper reasoning (*Hint: Make use of the pattern of reasoning employed in Question 1*).

1. Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
2. A convertible car is fun to drive. Isaac’s car is not a convertible. Therefore, Isaac’s car is not fun to drive.
3. Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
4. All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps.
5. All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.
6. Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.

7. All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
8. Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.